DE–6900 **13**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2013.

CLASSICAL ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Show that $\lim_{n\to\infty} \frac{3n^2+2n+5}{6n^2+4n+7} = \frac{1}{2}$.

   (b) Let $a_n = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n}$. Show that (a_n) coverges.

2. (a) (i) Test the convergence of the series

$$\frac{1}{3} + \frac{1.2}{3.5} + \frac{1.2.3}{3.5.7} + \cdots$$

   (ii) Test the convergence of the series $\sum \frac{2^n.n!}{n^n}$

   (b) Show that the series $\sum(-1)^n\left(\sqrt{n^2+1}-n\right)$.

3. (a) Find the greatest term in the expansion of $(1+x)^{\frac{13}{2}}$ when $x = \frac{2}{3}$.

   (b) Sum the series to infinity $\frac{15}{16} + \frac{15.21}{16.24} + \frac{15.21.27}{16.24.32} + \cdots$

4. (a) Sum the series

$$\frac{1^2}{1!}+\frac{1^2+2^2}{2!}+\frac{1^2+2^2+3^2}{3!}+\cdots+\frac{1^2+2^2+\cdots+n^2}{n!}+\cdots$$

(b) Show that

$$\log\sqrt{12}=1+\left(\frac{1}{2}+\frac{1}{3}\right)\frac{1}{4}+\left(\frac{1}{4}+\frac{1}{5}\right)\frac{1}{4^2}+\left(\frac{1}{6}+\frac{1}{7}\right)\frac{1}{4^3}+\cdots$$

5. (a) If α,β,γ are the roots of $x^3+px^2+qx+r=0$ form the equation whose roots are $\beta+\gamma-2\alpha$, $\gamma+\alpha-2\beta$ and $\alpha+\beta-2\gamma$.

(b) Solve the equation $x^4+20x^3-143x^2+430x+462=0$ by removing its second term.

6. (a) Find the condition that the equations $ax^3+3bx+c=0$, $a'x^3+3b'x+c'=0$ should have a common root. When this condition is satisfied, show that the common root is a double root of the equation $2(ab'-a'b)x^3+(ac'-a'c)x^2+(bc'-b'c)=0$.

(b) If α,β,γ are the roots of the equation $x^3+px^2+qx+r=0$, find the values of (α^2+1) (β^2+1) (γ^2+1).

7. (a) Find the inverse of the matrix $\begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 4 \\ -2 & 1 & 1 \end{bmatrix}$.

(b) Solve by Cramer's rule $x+y+z=0$, $2x+y-z=-7$, $3x+y-2z=-11$.

8. (a) Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.

(b) Show that the system of equations $x + 2y + z = 11$, $4x + 6y + 5z = 8$ and $2x + 2y + 3z = 19$ is consistent.

DE–6901 **14**

DISTANCE EDUCATION

B.Sc.(Mathematics) DEGREE EXAMINATION,
DECEMBER 2013.

CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Differentiate $\log(\sin x)$ and $\sin^{-1}\left(\frac{2x}{1-x^2}\right)$ with respect to x.

   (b) If $y = \log\sqrt{\frac{1+\sin x}{1-\sin x}}$, prove that $\frac{dy}{dx} = \sec x$.

2. (a) From any point of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, perpendiculars are drawn to the coordinate axes. Prove that the envelope of the straight lines joining the feet of these perpendiculars is the curve $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{a}\right)^{2/3} = 1$.

   (b) Show that the radius of curvature at any point of the catenary $y = c\cosh\left(\frac{x}{c}\right)$ is y^2/c.

3. (a) Evaluate $\int \frac{2x-1}{5x^2-x+2}dx$.

(b) Evaluate $\int \frac{\sec^2 x}{\sqrt{9+\tan^2 x}}dx$.

4. (a) Evaluate $\int_0^{\pi/2} \log(1+\tan x)\,dx$.

(b) If $I_n = \int_0^a x^n.e^{-x}dx$,

prove that $I_n - (n+a)I_{n-1} + a(n-1)I_{n-2} = 0$

5. (a) Solve $\frac{dy}{dx} - y\cot x = 2x\sin x$.

(b) Solve $xyp^2 + (3x^2 - 2y^2)p - 6xy = 0$.

6. (a) Solve $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} - 5y = \sin(\log x)$.

(b) Solve $(D^2 + 3D + 2)y = x^2$.

7. (a) Find $L(x^2\sin 2x)$.

(b) Find $L^{-1}\left[\frac{1}{s(s+1)(s+2)}\right]$.

8. (a) Solve $p\cot x + q\cot y = \cot z$.

(b) Solve $(p^2 + q^2)y = qz$.

DE–6902 **15**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2013.

ANALYTICAL GEOMETRY AND VECTOR CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Show that the angle between two diagonals of a cube is $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$.

   (b) Find the equation of the plane passing through the points $(1,1,1)$, $(1,2,1)$ and $(-2,2,-1)$.

2. (a) Prove that the straight line $5x+12y-4=0$ touches the circle $x^2+y^2-6x+4y+12=0$. Also find the point of contact.

   (b) Show that in a conic the semi - latus rectum is harmonic mean between the segments of a focal chord.

3. (a) Show that the lines $\frac{x-2}{1}=\frac{y-4}{2}=\frac{z-5}{2}$ and $\frac{x-5}{2}=\frac{y-8}{3}=\frac{z-7}{2}$ are coplanor and find the equation of the plane containing them.

   (b) Show that the planes $ax+hy+gz=0$, $hx+by+fz=0$ and $gx+fy+cz=0$ pass through a line if $abc+2fgh=af^2-bg^2-ch^2=0$.

4. (a) Find the equation of the sphere passing through the circle $x^2+y^2+z^2-4=0$, $2x+4y+6z-1=0$ and having its centre on the plane $x+y+z=6$.

(b) Find the equation of the spheres that passes through the two points $(0,3,0)$, $(-2,-1,-4)$ and cuts orthogonally the two spheres $x^2+y^2+z^2+x-3z-2=0$ and $2(x^2+y^2+z^2)+x+3y+4=0$.

5. (a) If $\vec{r}=\vec{a}\cos\omega t+\vec{b}\sin\omega t$ where $\vec{a}$ and $\vec{b}$ are constant vectors and ω is a constant, prove that $\vec{r}\times\frac{d\vec{r}}{dt}=\omega(\vec{a}\times\vec{b})$ and $\frac{d^2\vec{r}}{dt^2}+\omega^2\vec{r}=0$.

(b) Prove that curl (curl f) = grad div $f-\nabla^2 f$.

6. (a) Verify Green's theorem for the function $f=(x^2+y^2)\vec{i}-2xy\vec{j}$ and C is the rectangle in the $x-y$ plane bounded by $y=0$, $y=b$, $x=0$ and $x=a$.

(b) Evaluate $\int_c (e^x dx+2ydy-d2)$ by using stoke's theorem where C is the curve $x^2+y^2=4$, $z=2$.

7. (a) Find the condition that the lines

$a_1x+b_1y+c_1z+d_1=0=a_2x+b_2y+c_2z+d_2$ and

$a_3x+b_3y+c_3z+d_3=0=a_4x+b_4y+c_4z+d_4$ may be coplanor.

(b) Find in symmetrical form, the equation of the line given by $x+5y-z=7$, $2x-5y+3z+1=0$.

8. (a) Find the equation of the plane passing through the given points $(2,2,1)$ and $(9,3,6)$ and perpendicular to the plane $2x+6y+6z=9$.

(b) Obtain the equation of the sphere having the circle $s=x^2+y^2+z^2-3x+4y-2z-5=0$; $\pi=5x-2y+4z+7=0$ as a great circle.

DE–6903 **23**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2013.

MECHANICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) If three forces acting at a point can be represented in magnitude and direction by the sides of a triangle taken is order, prove that they will be in equillibrium. (10)

   (b) Two forces act on a particle. If the sum and difference of the forces are at right angles to each other, show that the forces are of equal magnitude. (10)

2. (a) If forces $\lambda\overline{OA}$ and $\mu\,\overline{OB}$ act a point O along the line OA and OB, prove that their resultant will be the force $(\lambda+\mu)\,\overline{OC}$ where C is the point on AB such that $\lambda.\ AC = \mu.\ CB$. (10)

   (b) P is a point in the plane of the triangle ABC and I is the incentre. Show that the resultant of forces represented by $PA.\sin A$, $PB.\sin B$, and $PC.\sin C$ is 4 PI. $\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$. (10)

3. (a) Find the resultant of two like parallel forces acting on a rigid body. (10)

(b) State and prove Varigon's theorem of moments. (10)

4. (a) Three like parallel forces acting at the vertices of a triangle, have magnitudes proportional to the opposite sides. Show that their resultant passes through the incentre of the triangle. (10)

(b) The resultant of three forces P, Q, R acting along the sides BC, CA, AB of a triangle ABC passes through the orthocentre. Show that the triangle must be obtuse angled. If $\angle A = 120^\circ$ and $B = C$, show that $Q + R = P\sqrt{3}$. (10)

5. (a) If the greatest height attained by the particle is a quarter of its range on the horizontal plane through the point of projection, find the angle of projection. (10)

(b) A particle is thrown over a triangle from one end of a horizontal base and grazing the vertex falls on the other end of the base. If A, B are the base angles and α the angle of projection, show that $\tan\alpha = \tan A + \tan B$. (10)

6. (a) 8 centimeters of rain fall in a certain district in 24 hours. Assuming that the drops fall freely from a height of 109 meters, find the pressure on the ground per square kilometer of the district. (10)

(b) A particle falls from a height h upon a fixed horizontal plane. If e is the coefficient of restitution, show that the whole distance described before the particle has finished rebounding is $h = \left(\frac{1+e^2}{1-e^2}\right)$ show also that the whole time taken is $\frac{1+e}{1-e}\sqrt{\frac{2h}{g}}$. (10)

7. (a) Show that the energy of a system executing S.H.M. is proportional to the square of the amplitude and of the frequency. (10)

(b) Show that the resultant of two simple harmonic motions in the same direction and of equal periodic time, the amplitude of one being twice that of the other and its phase a quarter of a period in advance, is a simple harmonic motion of amplitude $\sqrt{5}$ times that of the first and whose phase is in advance of the first by $\frac{\tan^{-1} 2}{2\pi}$ of a period. (10)

8. (a) Discuss the composition of two simple Harmonic motions of the same period in two perpendicular directions. (10)

(b) A particle moves in a phase with an acceleration which is always directed to a fixed point O in the plane. Obtain the differential equation of its path. (10)

DE–6904 **24**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2013

ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) (i) Define metric space. (2)

(ii) Let M be the set of all sequences in R. Let $x, y \in M$ and Let $x = (x_n)$ and $y = (y_n)$. Define $d(x,y) = \sum_{n=1}^{\infty} \frac{|x_n - y_n|}{2^n(1+|x_n - y_n|)}$. then prove that d is a metric on M. (8)

(b) Let M be a metric space and M_1 be a subspace of M. Let $A_1 \subseteq M_1$. Then prove that A_1 is open in M_1 iff there exists an open set A in M such that $A_1 = A \cap M_1$. (10)

2. (a) Let (M,d) be a metric space. Let $A, B \subseteq M$. Then prove that,

(i) $A \subseteq B \Rightarrow Int\, A \subseteq Int\, B$.

(ii) $Int(A \cap B) = Int\, A \cap Int\, B$.

(iii) $Int(A \cup B) \supseteq Int\, A \cup Int\, B$. (10)

(b) Let (M,d) be a metric space. Let $A,B \subseteq M$. Then prove that

(i) $A \subseteq B \Rightarrow \overline{A} \subseteq \overline{B}$.

(ii) $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

(iii) $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$. (10)

3. (a) Let M be a metric space and $A \subseteq M$. Then prove that $\overline{A} = A \cup D(A)$. (10)

(b) State and prove Baire category theorem. (10)

4. (a) Let (M_1,d_1) and (M_2,d_2) be 2 metric spaces. Prove that $f : M_1 \to M_2$ is continuous iff $f^{-1}(G)$ is open in M_1 whenever G is open in M_2. (10)

(b) Let $f : [a,b] \to \mathbf{R}$ be a monotonic increasing function. Then prove that f has a left limit and a right limit at every point of (a,b). Also prove that f has a right limit at a and f has a left limit at b. Further $x < y \Rightarrow f(x+) \leq f(y-)$. (10)

5. (a) Prove that a subspace of $\mathbf{R}$ is connected if and only if it is an interval. (10)

(b) Prove that a closed subspace of a compact metric space is compact. (10)

6. (a) Show that a metric space M is compact if and only if any family of closed sets with finite intersection property has non-empty intersection. (10)

(b) Let f be a continuous map from a compact metric space M_1 into any metric M_2. Then prove that $f(M_1)$ is closed and bounded. (10)

7. (a) Let $f_n; [0,1] \to \mathbf{R}$ be defined by $f_n(x) = \frac{x}{n}$. Prove that the sequence (f_n) converges pointwise to $f : [0,1] \to \mathbf{R}$ defined by $f(x) = 0$. (10)

(b) State and prove Cauchy criterion for uniform convergence. (10)

8. (a) State and prove Picards existence theorem. (16)

(b) Define contraction mapping with an example. (4)

DE–6905 **25**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2013.

PROBABILITY AND STATISTICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) If A and B are any two events of a sample space S then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

   (b) State and prove Baye's theorem.

2. (a) A coin is tossed as many times as is necessary to turn up head. Thus the elements of the sample space S are *H, TH, TTH, TTTH, TTTTH,* etc. We assign to these elements the respective probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$ etc. Show that this defines a probability set function P. Let

   $A = \{H, TH, TTH, TTTH, TTTTH\}$ and

   $B = \{TTTTH, TTTTTH\}$.

   Compute

   (i) $P(A)$.

   (ii) $P(B)$.

   (iii) $P(A \cap B)$.

   (iv) $P(A \cup B)$.

(b) If A and B are two events with $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cap B) = 0.2$. Find the probability of the events

(i) $A \cup B$.

(ii) $\overline{A} \cup \overline{B}$.

(iii) $\overline{A} \cap B$.

3. (a) A random variable X has the following probability function

x_i :	–2	–1	0	1	2	3
P_i :	0.1	K	0.2	$2K$	0.3	K

Find :

(i) The value of K.

(ii) Mean.

(iii) Variance.

(iv) $P(X \geq 2)$.

(v) $P(X < 2)$.

(vi) $p(-1 < X < 3)$.

(b) A continuous random variable X is defined by

$$f(x) = \begin{cases} \frac{1}{16}(3+x)^2 & \text{if } -3 \leq x < -1 \\ \frac{1}{16}(6-2x)^2 & \text{if } -1 \leq x < 1 \\ \frac{1}{16}(3-x)^2 & \text{if } -1 \leq x \leq 3. \end{cases}$$

Verify that

(i) the area under the curve is unity.

(ii) also show that the mean is zero.

4. (a) Obtain the

(i) mean

(ii) median and

(iii) mode

for the following distribution

$$f(x) = \begin{cases} 6(x - x^2) & \text{if} \quad 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

(b) A random variable X has the following probability function :

x :	1	2	3	4	5	6	7
$P(x)$:	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

(i) Find K.

(ii) Find $P(X \geq 6)$.

(iii) $P(X < 6)$.

(iv) $P(1 \leq X \leq 5)$.

(v) $E(X)$.

5. (a) Fit a second degree parabola by taking x_i as the independent variable.

x :	0	1	2	3	4
y :	1	5	10	22	38

(b) Find the correlation coefficient for the following data :

x :	10	12	18	24	23	27
y :	13	18	12	25	30	10

6. (a) Find the rank correlation coefficient between the height in cm and weight in kg of 6 soldiers in Indian Army.

Height :	165	167	166	170	169	172
Weight :	61	60	63.5	63	61.5	64

(b) Prove that the correlation coefficient lies between -1 and $+1$.

7. (a) An insurance agent accepts policies of 5 men all of identical age and in good health. The probability that a man of this age will be alive 30 years. Hence is 2/3. Find the probability that in 30 years.

(i) all five men.

(ii) atleast one man.

(iii) atmost three will be alive.

(b) Assuming that one is 80 births in a case of twins calcualte the probability of 2 or more births of twins on a day when 30 births occur using

(i) binomial distribution.

(ii) Poisson approximation.

8. (a) The theory predicts that the proportion of an object available in four groups *A, B, C, D* should be 9 : 3 : 3 : 1. In an experiment among 1600 items of this object the numbers in the four groups were 882, 313, 287 and 118. Use r^2 – test to verify whether the experimental result supports the theory.

(b) Fit a Poisson distribution for the following data and test the goodness of fit.

x :	0	1	2	3	4	5	6	Total
f :	273	70	30	7	7	2	1	390

DE–6906 **31**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2013.

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) It is found that among middle aged women 65% read Ananda Vikatan whereas, 76% read Kumudam. Find the percentage of middle ages women who read both. (5)

   (b) If A and B are two finite sets then prove that $|A \cup B| = |A| + |B| - |A \cap B|$. (5)

   (c) Let ρ be an equivalence relation defined on a set S. Then prove that

   (i) $a \rho b \Leftrightarrow [a] = [b]$

   (ii) Any two distinct equivalence classes are disjoint

   (iii) S is the union of all the equivalence classes. (10)

2. (a) Let $f: A \to B$, $g: B \to c$ be bijections. Then prove that $g \circ f: A \to C$ is also a bijection. (4)

   (b) Show that $f: R - \{3\} \to R - \{1\}$ given by $f(x) = \dfrac{x-2}{x-3}$ is a bijection and find its inverse. (6)

   (c) If ρ and σ are equivalence relations defined on a set s, prove that $\rho \cap \sigma$ is an equivalence relation. (10)

3. (a) Let G be the set of all real number except -1. Define $*$ on G by $a*b=a+b+ab$. Then prove that $(G, *)$ is a group. (5)

(b) If H and K are subgroups of a group G then prove that $H \cap K$ is also a subgroup of G. (5)

(c) Let H and K be two subgroups of a finite group G such that $|H| > \sqrt{|G|}$ and $|K| > \sqrt{|G|}$. Then prove that $H \cap K \neq \{e\}$. (10)

4. (a) Prove that isomorphism is an equivalence relation among groups. (10)

(b) Let G be any group. Show that $f: G \to G$ given by $f(x) = x^{-1}$ is an isomorphism $\Leftrightarrow G$ is abelian. (5)

(c) Let $f: G \to G'$ be a homomorphism. Then prove that the kernel K of f is a normal subgroup of G. (5)

5. (a) If R is a ring such that $a^2 = a$ for all $a \in R$, prove that

(i) $a + a = 0$

(ii) $a + b = 0 \Rightarrow a = b$

(iii) $ab = ba$. (10)

(b) Prove that a finite commutative ring without zero divisors is a field. (5)

(c) In an integral domain D of characteristic p, prove that the order of every element in the additive group is p. (5)

6. (a) Let R and R' be rings and $f: R \to R'$ be an isomorphism. Then prove that

   (i) R is commutative $\Rightarrow R'$ is commutative

   (ii) R is ring with identity $\Rightarrow R'$ is a ring with identity. (10)

   (b) Let R be a ring with identity. Then prove that $s = \{n \cdot 1 / n \in \mathbf{Z}\}$ is a subring of R. (5)

   (c) Prove that $R[x]$ is an integral domain iff R is an integral domain. (5)

7. (a) If W is any subspace of V containing A and B then prove that W contains $A + B$. (10)

   (b) Prove that any subset of a linearly independent set is linearly independent. (5)

   (c) Let V be a finite dimensional vector space over a field F. Prove that any linearly independent set of vectors in V is part of a basis. (5)

8. (a) State and prove Schwartz's inequality. (5)

   (b) Apply Gram-Schmidt process to construct an orthonormal basis for $V_3(R)$ with the standard inner product for the basis $\{v_1, v_2, v_3\}$ whereas $v_1 = (1, 0, 1)$; $v_2 = (1, 3, 1)$ and $v_3 = (3, 2, 1)$. (10)

   (c) Prove that $V = W \oplus W^{\perp}$. (5)

DE–6907 **32**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2013.

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Solve graphically the LPP :

Maximize $Z = 3x_1 + 2x_2$

Subject to : $5x_1 + x_2 \geq 10$

$x_1 + x_2 \geq 6$
$x_1 + 4x_2 \geq 12,$

$x_1, x_2 \geq 0$. (10)

(b) If $x_1 = 2, x_2 = 3, x_3 = 1$ be a feasible solution of the following linear programming problem then find a basic feasible solution;

Maximize $z = x_1 + 2x_2 + 4x_3$

Subject to the constraints :

$2x_1 + x_2 + 4x_3 = 11$
$3x_1 + x_2 + 5x_3 = 14,$

$x_1, x_2, x_3 \geq 0$. (10)

2. (a) Three grades of coal A, B and C contain phosphorus and ash as impurities. In a particular industrial process, fuel up to 100 ton (maximum) is required which should contain ash not more than 3% and phosphorus not more than 0.03%. It is a desire to maximize the profit while satisfying these conditions. There is an unlimited supply of each grade. The percentage of impurities and the profits of grades are given below :

Coal	Phosphorus (%)	Ash(%)	Profit in Rs. (per ton)
A	0.02	3.0	12.00
B	0.04	2.0	15.00
C	0.03	5.0	14.00

Find the proportions in which the three grades be used, to maximum the profit. (10)

(b) Using Simplex method, solve the following LPP :

Minimize $Z = x_1 - 3x_2 + 2x_3$,

Subject to : $3x_1 - x_2 + 2x_3 \le 7$,

$-2x_1 + 4x_2 \le 12$,

$-4x_1 + 3x_2 + 8x_3 \le 10$,

$x_1, x_2, x_3 \ge 0$. (10)

3. (a) Use the method of penalty to solve the following LPP :

Maximize $Z = 2x_1 + 3x_3$

Subject to : $x_1 + x_2 + 2x_3 \le 5$,

$2x_1 + 3x_2 + 4x_3 = 12$,

$x_1, x_2, x_3 \ge 0$. (10)

(b) Use two – phase simplex method to

Minimize $z = x_1 + x_2$

Subject to : $2x_1 + x_2 \ge 4$,

$x_1 + 7x_2 \ge 7$,

$x_1, x_2 \ge 0$. (10)

4. (a) Use Charnes penalty method to solve the LPP :

Maximize $z = x_1 + 2x_2 + 3x_3 - x_4$

Subject to : $x_1 + 2x_2 + 3x_3 = 15$

$2x_1 + 2x_2 + 5x_3 = 20,$

$x_1 + 2x_2 + x_3 + x_4 = 10$ (10)

$x_1, x_2, x_3, x_4 \geq 0.$

(b) Find the optimum integer solution to the LPP :

Maximize $z = 3x_2$

Subject to : $3x_1 + 2x_2 \leq 7,$

$x_1 + x_2 \geq -2,$

$x_1, x_2 \geq 0$ and are integers. (10)

5. (a) Explain North – West Corner method to find initial basic feasible solution to a LPP. (10)

(b) Find the optimum solution to the following Transportation problem : (10)

Warehouse

Factory	D	E	F	G	Capacity
A	42	48	38	37	160
B	40	49	52	51	150
C	39	38	40	43	190
Demand	80	90	110	160	

6. Solve the transportation problem with transportation costs in rupees, requirements and availability as given below : (20)

	D	D	D	D	a_i
S_1	10	15	12	12	200
S_2	8	10	11	9	150
S_3	11	12	13	10	120
b_j	140	120	80	220	560/470

7. (a) Determine which of the following two-person zero-sum are strictly determinable and fair. Give optimum strategies for each player in the case of strictly determinable games :

(i) $$\begin{array}{cc} & \text{Player B} \\ \text{Player A} & \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} \end{array}$$

(ii) $$\begin{array}{cc} & B \\ A & \begin{bmatrix} 0 & 2 \\ -1 & 4 \end{bmatrix} \end{array}.$$ (10)

(b) For the game with the following payoff matrix, determine the optimum strategies and the value of the game : (10)

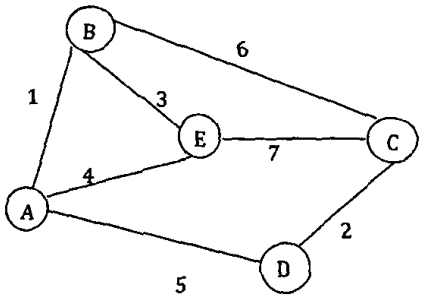
$$\begin{array}{cc} & P_2 \\ P_1 & \begin{pmatrix} 0 & 2 \\ -1 & 4 \end{pmatrix} \end{array}$$

8. (a) Construct the network diagram comprising activities B, C, ... Q and N such that the following constrains are satisfied :

B<EF: C<G.L: E.G<H: FG <H: L.H<l:

L<M: H.M<N: H< J: I.J< P: P<Q. (10)

(b) Use Dijkstra's algorithm to determine a shortest path from city A to city C for the following network : (10)



DE–6908 **33**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2013.

NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Using Newton-Raphson method to find the real root of $x \log_{10}^{x} = 1.2$ correct to four decimal places.

2. Solve the equation $x + y = 2$ and $2x + 3y = 5$ by

   (a) Gauss elimination method

   (b) Gauss-Jordan elimination method.

3. Apply Gauss forward interpolation formula to find $y(25)$ for the following data.

x	20	24	28	32
y	2854	3162	3544	3992

4. Derive Lagrange's interpolation formula and using this formula to find the value of y at $x = 6$ from the following data.

x	3	7	9	10
y	168	120	72	63

5. Apply Newton's forward interpolation formula to find $y'(x)$ given

x	0	1	2	3	4
$y(x)$	1	1	15	40	85

Hence find $y'(x)$ at $x = 0.5$.

6. Evaluate $\int_0^2 \frac{dx}{x^2+4}$ using Romberg's method. Hence obtain an approximate value of π.

7. Find the successive approximate solution of the differential equation $y' = y$, $y(0) = 1$ by Picard's method and compare it with the exact solution.

8. Using Taylor's method to solve $\frac{dy}{dx} = 1 + xy$ with $y(0) = 2$ find

(a) $y(0.1)$

(b) $y(0.2)$ and

(c) $y(0.3)$.

DE–6909 **34**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2013

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Let $f(z) = u + iv$ be a function defined in a region D such that u, v and their first order partial derivatives are continuous in D. If the first order partial derivatives of u, v satisfy the C-R equations at a point $(x, y) \in D$ then prove that f is differentiable at $z = x + iy$.

   (b) State and prove C-R equations in polar co-ordinates.

2. (a) Prove that the function $f(z) = \begin{cases} \dfrac{x^3(1+i) - y^3(1-i)}{x^2 + y^2} & if\ z \neq 0 \\ 0 & if\ z = 0 \end{cases}$ satisfies C-R equations at the origin but $f'(0)$ doe not exist. (5)

   (b) If $\dfrac{\partial^2}{\partial x \partial y} = \dfrac{\partial^2}{\partial y \partial x}$ then prove that $\dfrac{\partial^2}{\partial x^2} = \dfrac{\partial^2}{\partial y^2} = 4\dfrac{\partial^2}{\partial z \partial \bar{z}}$. (10)

   (c) The real and imaginary parts of an analytic function are harmonic functions. (5)

3. (a) Find the image of the strip $2 < x < 3$ under $w = \frac{1}{z}$.

(b) Define mobius transaformation and prove that any bilinear transformation can be expressed as a product of translation, rotation, magnification or contraction and inversion.

4. (a) Determine the bilinear transformation which maps $0,1,\infty$ in to $i,-1,-i$ respectively. Under this transformation show that the interior of the unit circle of the z-plane maps onto the unit circle of the z-plane maps onto the half plane left to the v-aixs (left half of the w -plane). (10)

(b) Prove that a bilinear transformation preserves inverse points. (5)

(c) Find the invariant points of the transformation $w = \frac{1+z}{1-z}$. (5)

5. (a) Prove that $\int_c \frac{dz}{(z-a)^n} = \begin{cases} 0 & if\, n \neq 1 \\ 2\pi i & if\, n = 1 \end{cases}$ where C is the circle with centre a and radius r and $n \in Z$.

(b) State and prove Cauchy's theorem.

6. (a) Let f be analytic inside and on a simple closed curve C. Let z be any point incide C. Then $f'(z) = \frac{1}{2\pi i}\int_c \frac{f(s)}{(s-z)^2} dz$.

(b) Evaluate $\int_c \frac{e^z}{(z+2)(z+1)^2} dz$ where C i $|z| = 3$.

7. (a) If $f(z)=\dfrac{z+4}{(z+2)(z+1)^2}$ find laurent's series expansion in (i) $0<|z-1|<4$ and (ii) $|z-1|>4$.

(b) Suppose $f(z)$ is analytic in a region D and is not identically zero in D. Then prove that the set of all zero's of $f(z)$ is isolated.

8. (a) Show that the function $2+z^2-e^{iz}$ ha precisely one zero in the open upper half plane.

(b) Use contour integration technique to find the value of $\displaystyle\int_0^{2\pi}\frac{d\theta}{2+\cos\theta}$.

DE–6910 **35**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2013.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Determine which of the following compound propositions are tautologies and which of them are contradictions, using truth tables.

(i) $\left(\neg q \wedge (p \to q)\right) \to \neg p$

(ii) $((p \to q) \wedge (q \to r)) \to (p \to r)$

(iii) $\neg(q - r) \wedge r \wedge (p \to q)$

(iv) $((P \vee q) \wedge (p \to r) \wedge (q \to r)) \to r$.

(b) Without constructing the truth tables, prove the following.

(i) $\neg p \to (q \to r) \equiv q \to (p \vee r)$

(ii) $p \to (q \to r) \equiv p \to \left(\neg q \vee r\right) \equiv (p \wedge q) \to r$

(iii) $\left((p \vee q) \wedge \neg\left(\neg p \wedge \left(\neg p \vee \neg r\right)\right)\right) \vee$

$\left(\neg p \wedge \neg q\right) \vee \left(\neg p \wedge \neg r\right)$

is a tautology.

2. (a) Show that $(p \to q) \wedge (r \to q)$ and $(p \vee r) \to q$ are equivalent formulae.

(b) Find the PDNF and PCNF of the following statements using truth tables.

(i) $p \to (p \wedge (q \to p))$

(ii) $(q \to p) \wedge (\neg p \wedge q)$

(iii) $(p \wedge q) \vee (p \wedge r) \vee (q \wedge r)$

(iv) $(p \wedge q) \vee (\neg p \wedge q) \vee (q \wedge r)$

3. (a) Show that $(p \to q) \wedge (r \to s), (q \to t) \wedge (s \to u)$, $\neg(t \wedge u)$ and $(p \to r) \Rightarrow \neg p$.

(b) Define an open statement. Explain through an example. Identify pronouns, objects and universe.

4. (a) Show that the premises "One student in this case knows how to write programs in JAVA" and "Everyone who knows how to write programs in JAVA can get a high-paying job" imply the conclusion "Someone in this class can get a high– paying job."

(b) Show that the conclusion $\forall x(P(x) \to \neg Q(x))$ follows from the premises. $\exists x(P(x) \wedge Q(x)) \to \forall y(R(y) \to S(y))$ and $\exists y(R(y) \wedge \neg S(y))$

5. (a) Prove that the number of vertices of odd degree in an undirected graph is even.

(b) The incidence matrices of two pairs of graphs are as follows. Examine the isomorphism of G and H either graphically or by finding a permutation matrix.

(i) $$I_G \equiv \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}; \; I_H \equiv \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

(ii) $$I_G \equiv \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix};$$

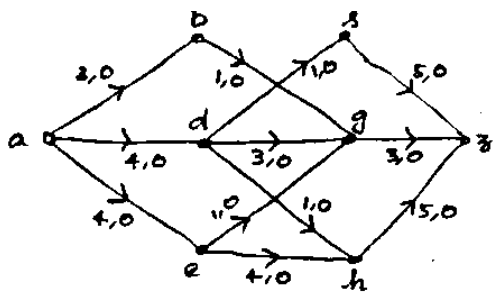
$$I_H \equiv \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

6. (a) Prove that an undirected graph is a tree, if and only if, there is a unique simple path between every pair of vertices.

(b) Sketch the II–vertex binary trees with minimum and maximum heights. Find also the path length of both the trees.

7. (a) Prove that a connected graph G is Eulerian if and only if it can be decomposed into cycles.

(b) Use the labeling procedure to find a flow for the transport network in the following figure.



8. (a) For a graph G, show that the vertex connectivity $\leq$ edge connectivity $\leq$ minimum degree δ. (6)

(b) Define

(i) Transport network

(ii) Saturated

(iii) Value of the flow. (6)

(c) If f is a flow in a transport network $N = (V, E)$, then prove that the value of the flow from the source a is equal to the value of the flow into the sink z. (8)